

An Analytical and Computational Survey of One-Dimensional Maps

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Abstract

This survey provides an analytical and computational exploration of one-dimensional iterated maps, serving as a mathematical laboratory for understanding the universal mechanics of chaos. By examining the discrete-time dynamics of the logistic and Ricker maps, the paper traces the transition from stable fixed points to chaotic regimes via the period-doubling route. The study employs foundational tools—including linear stability analysis, cobweb plots, and orbit diagrams—to visualize and analyze long-term attractors. Chaotic behavior is rigorously quantified through the numerical computation of the Lyapunov exponent to identify sensitive dependence on initial conditions. Furthermore, the analysis investigates the emergence of periodic windows from tangent bifurcations, specifically the appearance of a stable period-3 cycle within chaotic regions. Comparative results reveal profound qualitative and quantitative universality across unimodal maps, evidenced by the U-sequence and Feigenbaum’s constant. Ultimately, these findings demonstrate that the transition to chaos is governed by the fundamental geometry of the map rather than its specific algebraic formulation, offering significant implications for the study of higher-dimensional continuous dynamical systems.

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1 Introduction

As a kid, we've all played around with repeatedly pressing the equals button on a simple calculator, watching the results of the repeated $+1$'s slowly climb upward, or watching as the repeated $\times 2$'s quickly wound up at an overflow error. If you tried this with the cosine function, you might have noticed a fascinating outcome: the results of each subsequent calculation seem to converge to one number, somewhere around 0.739 [4]. The sequence of numbers that emerged from doing so can be defined by the rule $x_{n+1} = \cos x_n$, where x_0 is whatever number you began at. Such a rule is an example of a one-dimensional map, which is a dynamical system in which time is discrete and the numbers, x_n , belong to one dimension, the space of real numbers. One dimensional maps are also known as difference equations, recursion relations, iterated maps, or simply **maps**.

While our previous work in the course has primarily focused on 1-D and 2-D dynamical systems where time flows continuously, this paper explores this different class of systems where time is discrete. Despite this fundamental shift in how time is measured, we will see that these one-dimensional iterated maps exhibit many characteristics analogous to the continuous systems we have studied, including fixed points, stability thresholds, and bifurcations. Furthermore, these maps serve as highly effective yet simple mathematical laboratories for understanding complex chaotic behavior. By isolating and studying the dynamics of these discrete maps in their own right, we discover some profound insights into the universal mechanics of chaos that guide both discrete and continuous systems.

The general form of a map is $x_{n+1} = f(x_n)$, and the sequence of numbers generated by the map $x_0, x_1, x_2, \dots$ is called the **orbit** starting from x_0 . A **fixed point** of a map, denoted x^* , satisfies $f(x^*) = x^*$, for if $x^n = x^*$, then the orbit will remain at x^* for all future iterations.

This paper will primarily use **unimodal** maps where the functions $f(x)$ are smooth, concave down, and have a single maximum to explore the concepts covered. Specifically, the two maps to be explored are the **logistic map**,

$$x_{n+1} = rx_n(1 - x_n),$$

and the **Ricker map**,

$$x_{n+1} = x_n e^{r(1-x_n)},$$

which are both discrete time analogs of population growth models. In these two maps, $x_n \geq 0$ is a dimensionless measure of population and $r \geq 0$ is a variable intrinsic growth rate [4].

Additionally, there is some other terminology that we briefly introduce now and will define more rigorously later in the paper. The **multiplier** of a fixed point is defined as $\lambda = f'(x^*)$, and a fixed point is considered **linearly stable** (or just **stable**) if $|\lambda| < 1$, and **unstable** otherwise. We will explore orbits that eventually end up in a state of repeating every p iterations, known as **period- p cycles**, and track the parameter values r_n , where a **period- 2^n cycle** is first born. And to quantify chaos as sensitive dependence on initial conditions, we will use the **Lyapunov exponent**, denoted as λ .

The remainder of this paper is structured as follows. Section 2 establishes the foundational analytical and visual tools for one-dimensional discrete maps, focusing on fixed point stability criteria and cobweb diagrams. Section 3 presents a deep dive into the logistic map, examining its period-doubling route to chaos and the phenomenon of intermittency within periodic windows. To mathematically quantify this chaotic behavior, Section 4 introduces the Lyapunov exponent and presents numerical computations of its value across different parameter regimes. Section 5 broadens the scope by analyzing the Ricker map, utilizing it to demonstrate Feigenbaum's theory of qualitative and quantitative universality across different unimodal maps. Finally, Section 6 concludes with a synthesis of these discrete phenomena and their connections to the broader study of continuous dynamical systems.

2 Fixed Points, Stability, and Cobwebs

2.1 Fixed Points and Linear Stability

As mentioned earlier, $x = x^*$ is a fixed point if x^* satisfies $f(x^*) = x^*$. Then, if $x_n = x^*$, for all future iterations, $x_{n+1} = f(x_n) = f(x^*) = x^*$ and the orbit gets stuck at x^* . To determine the stability of a fixed point x^* , we consider a nearby orbit $x_n = x^* + \eta_n$ and determine whether η_n grows or decays as n increases. This will tell us whether the orbit is attracted to or repelled by x^* . Beginning from $x^* + \eta_{n+1}$, we apply substitution and a Taylor expansion to obtain

$$x^* + \eta_{n+1} = x_{n+1} = f(x^* + \eta_n) = f(x^*) + f'(x^*)\eta_n + O(\eta_n^2).$$

Because $f(x^*) = x^*$, we can subtract x^* from both sides to reduce the equation to

$$\eta_{n+1} = f'(x^*)\eta_n + O(\eta_n^2).$$

Dropping the higher order $O(\eta_n^2)$ terms, we arrive at $\eta_{n+1} = f'(x^*)\eta_n$. From this equation, we can see that $\eta_1 = \lambda\eta_0$, $\eta_2 = \lambda\eta_1 = \lambda^2\eta_0$, so, in general, $\eta_n = \lambda^n\eta_0$ where $\lambda = f'(x^*)$ is the **multiplier**. A fixed point was defined as **linearly stable** if $|\lambda| = |f'(x^*)| < 1$, as then, $\eta_n \rightarrow 0$ as $n \rightarrow \infty$, meaning nearby orbits are attracted by x^* . Conversely, if $|\lambda| = |f'(x^*)| > 1$, then $\eta_n \rightarrow \infty$ as $n \rightarrow \infty$, so nearby orbits are repelled by x^* . This is the **unstable** case.

To apply this analytical framework, we consider the logistic map, defined generally as $f(x) = rx(1-x)$. Let us examine the specific case where the growth rate parameter $r = 2.9$, which yields the map $f(x) = 2.9x(1-x)$. First, we must locate the fixed points by satisfying the condition $f(x^*) = x^*$:

$$2.9x^*(1-x^*) = x^*$$

This equation yields two solutions. The first is the trivial fixed point $x_1^* = 0$. Assuming $x^* \neq 0$, we can divide both sides by x^* to find the nontrivial fixed point:

$$2.9(1 - x_2^*) = 1 \implies x_2^* = 1 - \frac{1}{2.9} = \frac{19}{29}.$$

Next, we determine the linear stability of each fixed point by evaluating the multiplier $\lambda = f'(x^*)$ derived previously. The derivative of our specific map is given by: $f'(x) = 2.9(1 - 2x)$. For the trivial fixed point $x_1^* = 0$, the multiplier is $\lambda = f'(0) = 2.9$. Because $|\lambda| = 2.9 > 1$, the origin is unstable. In a biological context, this means that a small initial population will be repelled away from zero and grow rather than dying out. For the nontrivial fixed point $x_2^* = 19/29$, the multiplier evaluates to:

$$\lambda = f'\left(\frac{19}{29}\right) = 2.9 \left(1 - 2\left(\frac{19}{29}\right)\right) = -0.9.$$

Because $|\lambda| = |-0.9| = 0.9 < 1$, the fixed point $x_2^* = 19/29$ is linearly stable. Furthermore, because the multiplier λ is negative, the linear analysis tells us that nearby orbits will oscillate as they are attracted to x_2^* , meaning the population will alternate above and below the equilibrium value as it settles.

2.2 Cobwebs

Next, we introduce the **cobweb** plot for iterating a map visually. See the example generated in Python using NumPy and Matplotlib (Figure 1).

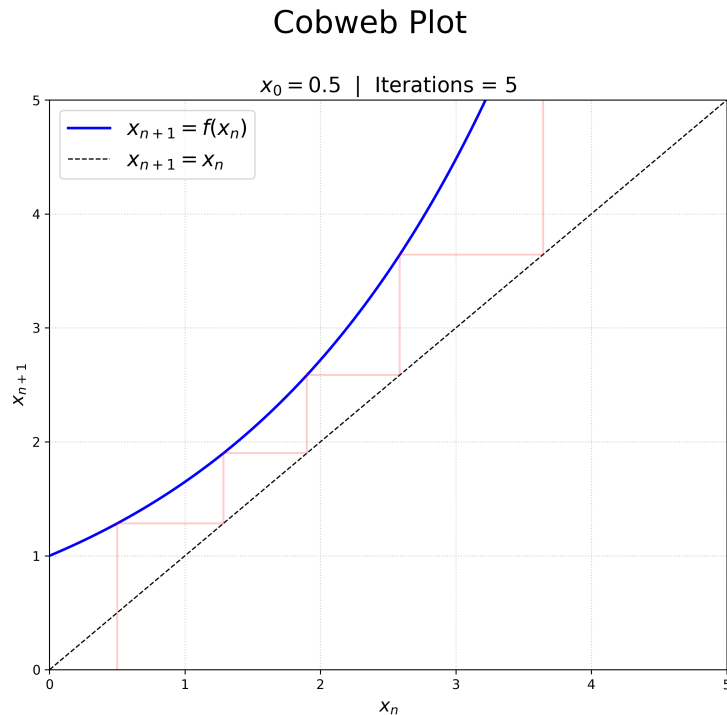


Figure 1: A simple cobweb plot where $x_{n+1} = e^{0.5x_n}$.

A graph where the axes are x_n and x_{n+1} , cobwebs are helpful tools for visualizing the global behavior and for determining stability in the **marginal** case, where $|f(x^*)| = 1$ and linearization does not yield any results. Given a map $x_{n+1} = f(x_n)$ and a starting condition x_0 , we begin the cobweb by drawing a vertical line up from the x_n axis until it intersects $f(x)$ at x_1 . Then, we trace a horizontal line from there until it intersects the diagonal, $x_{n+1} = x_n$. This process can be repeated n times to generate the first n points in the orbit starting from x_0 .

Again, let us apply this approach to the logistic map with $r = 2.9$ (Figure 2).

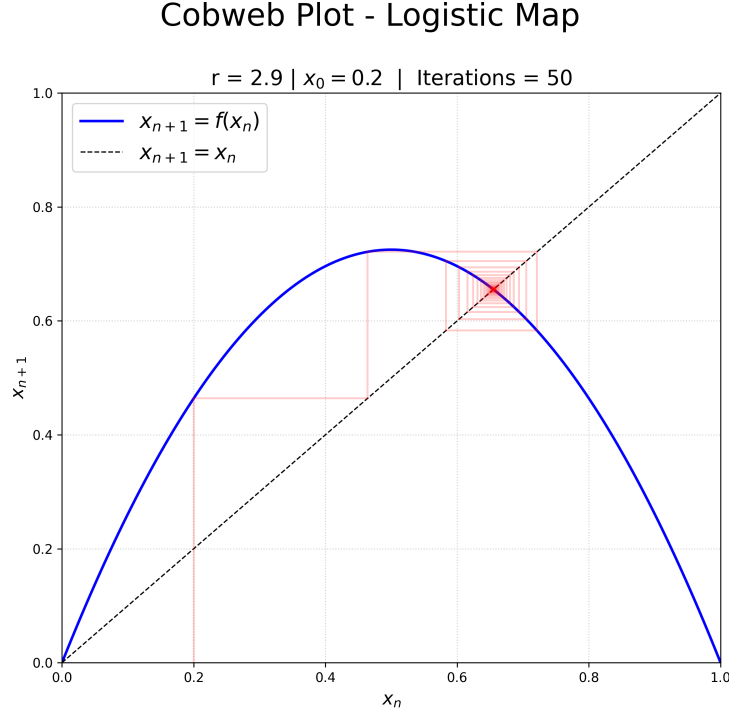


Figure 2: A cobweb plot for $x_{n+1} = 2.9x_n(1 - x_n)$.

This cobweb visually confirms the results we derived earlier for the stability of $x_2^* = 19/29 \approx 0.655$. In the cobweb, we can see that the orbit converges to this point. Furthermore, we can visually see the damped oscillations that occur as the orbit converges, just as predicted from $\lambda < 0$ in our linear stability analysis. The orbit oscillates alternatively above and below the equilibrium value as it converges.

3 Deep Dive: The Logistic Map

We now continue on a deeper dive into the logistic map.

3.1 Numerical and Analytical Results

For a small growth rate ($r < 1$), the population, as one might expect, always goes extinct as $x_n \rightarrow 0$ as $n \rightarrow \infty$. This can be confirmed with a cobweb diagram (Figure

3).

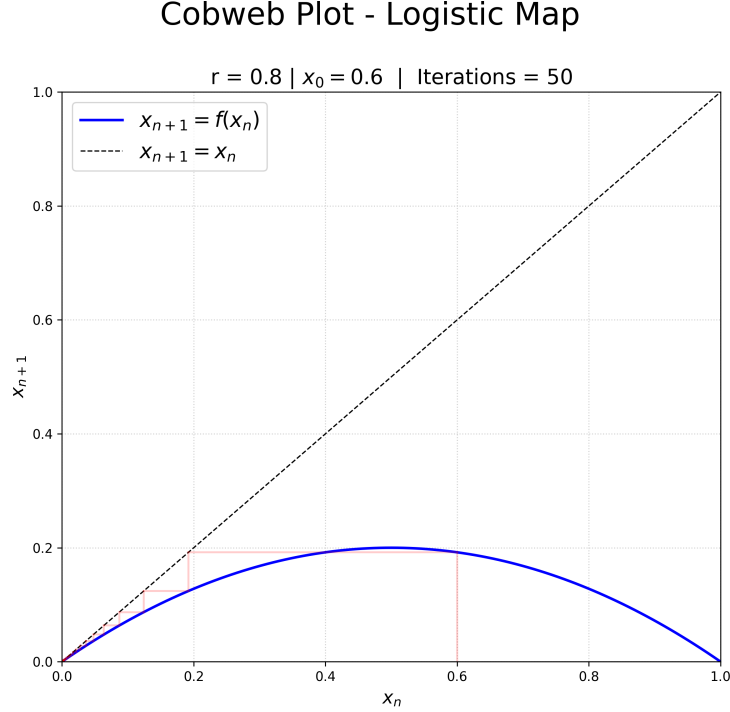


Figure 3: A cobweb plot for $x_{n+1} = 0.8x_n(1 - x_n)$.

For $1 < r < 3$, the population eventually converges to a steady state via damped oscillations, as seen in the earlier analytical and visual analysis for $r = 2.9$.

For a larger r like $r = 3.2$, we now try plotting the cobweb (Figure 4). Here, we see some interesting new behavior. The orbit no longer converges to a singular point, but rather seems to settle into an oscillation between two points: a period-2 cycle. Let us explore this result analytically. A 2-cycle exists if and only if there exists p and q such that $f(p) = q$ and $f(q) = p$ (so that $f(f(p)) = p$). This means that p is a fixed point of the **second-iterate map** $f(f(x))$, denoted as $f^2(x)$. Thus, finding the points of the 2-cycle is equivalent to finding the fixed points of $f^2(x)$, that is, solving $f^2(x) = x$. Expanding $f^2(x) - x$ gives us

$$r^2x(1-x)[1-rx(1-x)] - x = 0.$$

After factoring out the roots x and $x - (1 - \frac{1}{r})$, and solving for x , we obtain the fixed points p and q as follows:

$$p, q = \frac{r+1 \pm \sqrt{(r-3)(r+1)}}{2r}.$$

From this result, we can see that the roots are real for $r \geq 3$, so for $r < 3$, the roots are imaginary and the 2-cycle does not exist, just as we see in the analysis of the $r = 2.9$ case. Furthermore, at $r = 3$, we can see that

$$p, q = \frac{r+1 \pm \sqrt{(3-3)(r+1)}}{2r} = \frac{r+1}{2r} = \frac{2}{3}.$$

This coincides with the non-trivial fixed point of the logistic map occurring at $1 - \frac{1}{r} = 1 - \frac{1}{3} = \frac{2}{3}$. This shows that the 2-cycle bifurcates from x^* at $r = 3$.

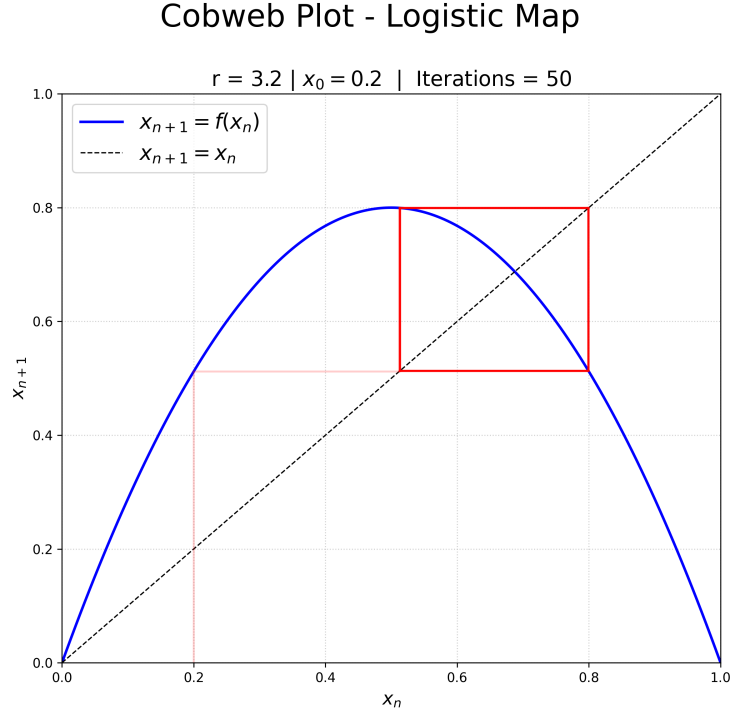


Figure 4: A cobweb plot for $x_{n+1} = 3.2x_n(1 - x_n)$.

If we check out a yet larger $r = 3.5$, the cobweb plot shows that now, the orbit settles into a period-4 cycle (Figure 5). Further increasing r to $r = 3.55$ further yields yet another **period-doubling** to a period-8 cycle (Figure 6). Note that the period- 2^n cycles are clearly visible in the cobwebs as the segments that are fully opaque (this was achieved by plotting the segments with stacked opacity, such that when the orbits settle into their cycles, they opacity stacks and clearly shows the segments that are a part of the cycles).

Cobweb Plot - Logistic Map

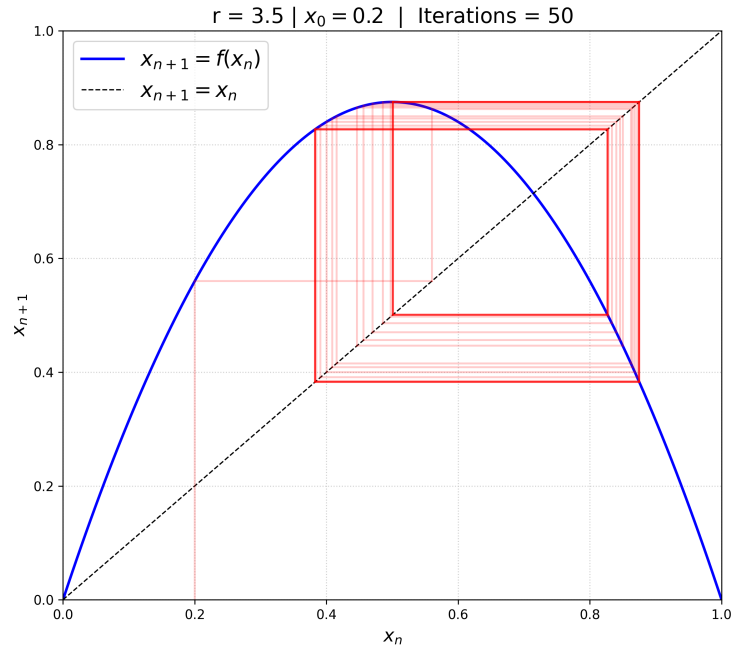


Figure 5: A cobweb plot for $x_{n+1} = 3.5x_n(1 - x_n)$.

Cobweb Plot - Logistic Map

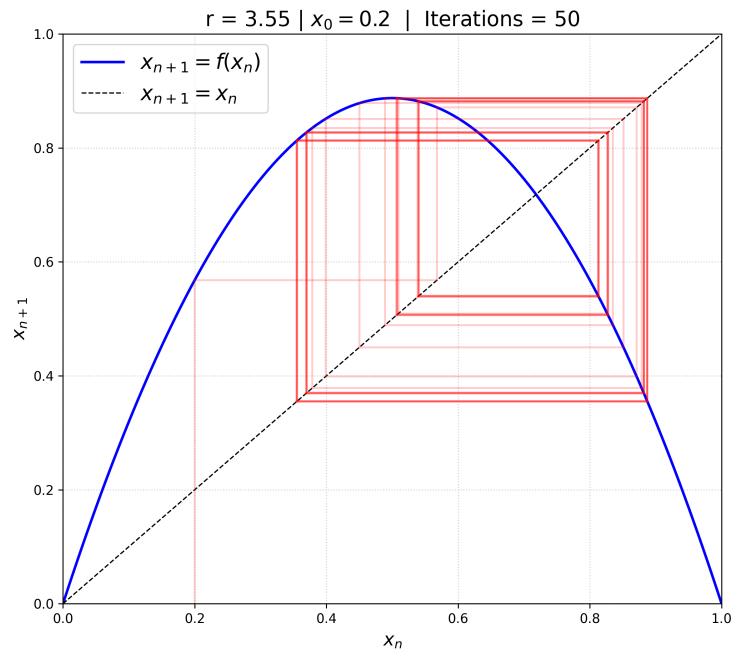


Figure 6: A cobweb plot for $x_{n+1} = 3.55x_n(1 - x_n)$.

Computer experiment results reveal that continued period-doublings occur as r increases. Here are the first few r_n , which are the values of r at which a 2^n -cycle first appears [4]:

$$\begin{aligned} r_1 &\approx 3 \\ r_2 &\approx 3.449 \\ r_3 &\approx 3.54409 \\ r_4 &\approx 3.5644 \\ &\dots \\ r_\infty &\approx 3.569946. \end{aligned}$$

The r_n 's ultimately converge to a limiting factor of r_∞ . At this point, the behavior of the system for $r > r_\infty$ becomes highly unpredictable. An accompanying cobweb also displays this (Figure 7).

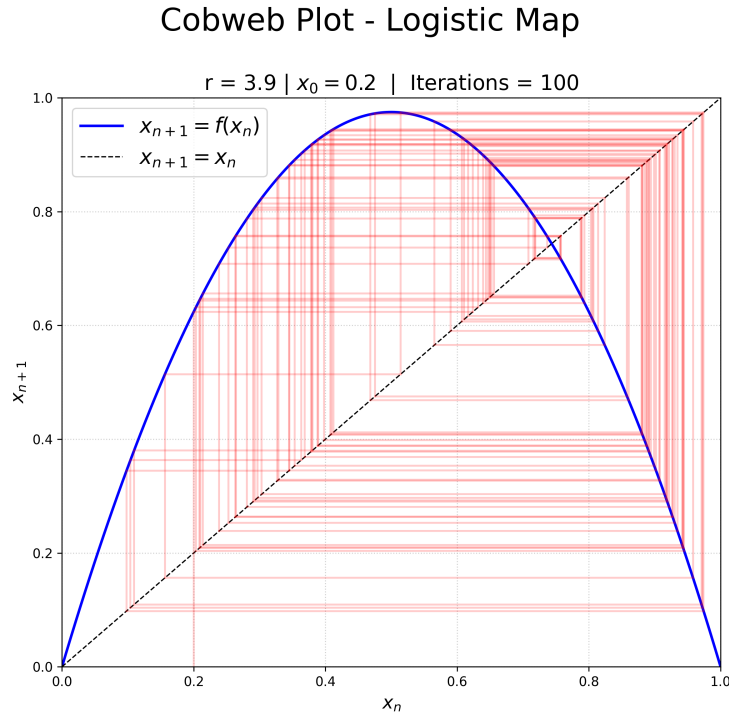


Figure 7: A cobweb plot for $x_{n+1} = 3.9x_n(1 - x_n)$.

For many values of $r > r_\infty$, the orbits never settle into a fixed point or a periodic cycle. This is a discrete-time version of chaos. In the next section, we will visualize this behavior with orbit diagrams.

3.2 Orbit Diagrams

We now want to investigate the long term effects of increasing r by visualizing orbits for all values of r at once. **Orbit diagrams** plot the attractor of a map as a function

of r . Attractors are the long term behavior of the orbit, so here's what this looks like computationally: for a given value of r , the map is iterated some number of times (i.e. 300) to allow for the orbit to settle into its attractor. At this point, the previous results are discarded and we plot the next k values of x_n for some large number of k (i.e. also 300) all on one vertical at the specific r value. Thus, the orbit diagram ends up displaying the points of the attractor (and not the early behavior of the orbit, as those points are discarded), as a function of r . Here's what this looks like for the logistic map, generated in Python (Figure 8):

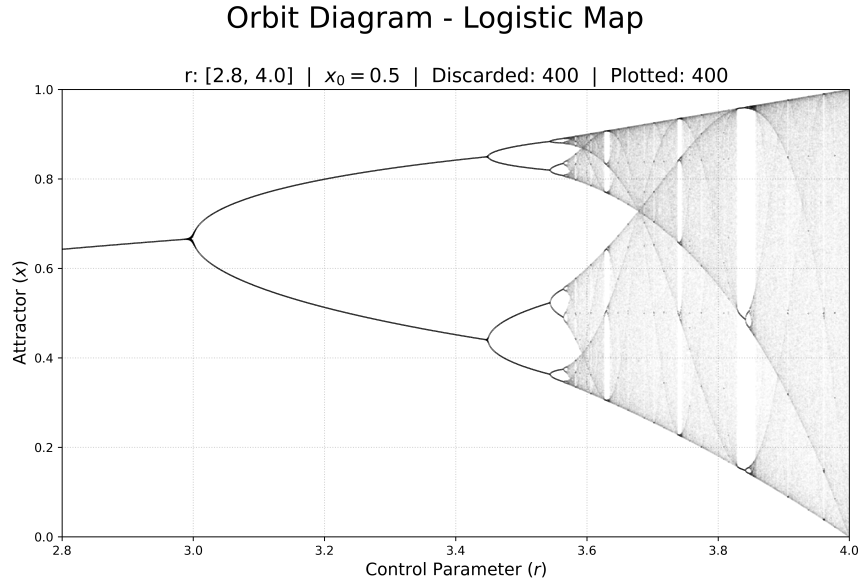


Figure 8: An orbit diagram for $x_{n+1} = rx_n(1 - x_n)$. This diagram has become famous within the field of nonlinear dynamics.

Here, the numerical results from earlier come to life. At $r_1 = 3$ the single fixed point $x^* = 1 - \frac{1}{r}$ splits into two branches, indicating the 2-cycle bifurcation we found analytically earlier. At successive r_n 's we see that each branch progressively splits into two more branches, displaying the period-doubling behavior we saw on the cobwebs. Finally, at $r_\infty \approx 3.569946$, we see that the orbits become chaotic and the attractors are no longer just a finite set of points. For $r > r_\infty$, the orbit diagram displays a curious mixture of both chaotic and periodic behavior. For example, starting around $r \approx 3.83$, we see that there is a rather large window containing a stable period-3 cycle that seemingly just appears out of the chaos. We will briefly explore this 3-cycle window in the next section.

3.3 Periodic Windows

In order to understand where this 3-cycle comes from, we must take a look at the third-iterate map $f^3(x) = f(f(f(x)))$ of the logistic map, just as we used the 2nd-iterate map in analyzing the 2-cycle. Analytical methods are no longer viable as $f^3(x)$ for the logistic map is an 8th degree polynomial. We will take a graphical approach instead,

as a simple graph of $f^3(x)$ is sufficient to explain the 3-cycle. First, let us look at the graph of $f^3(x)$ at an r value slightly before the 3-cycle window (Figure 9).

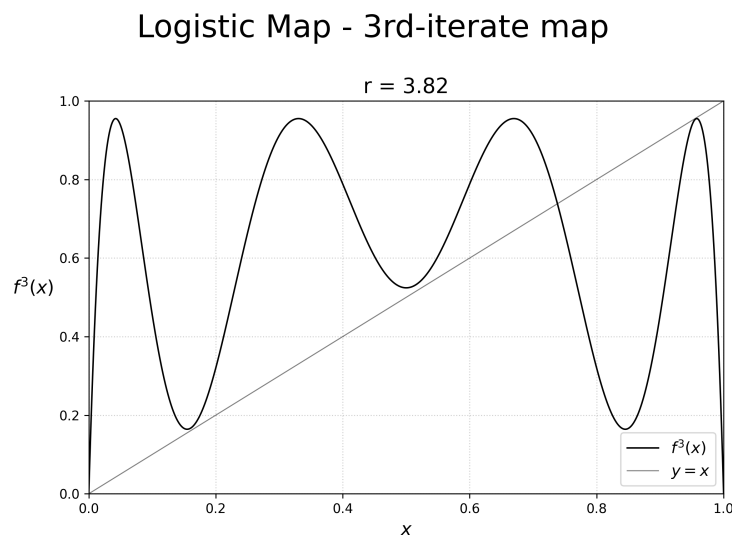


Figure 9: A graph of $f^3(x)$ for $x_{n+1} = 3.82x_n(1 - x_n)$.

These results are what we might expect: there is only one fixed point for the third-iterate map at this point, which is $x^* = 1 - \frac{1}{r}$, the fixed point of the logistic map itself. This makes sense—the period-3 cycle has not yet appeared at $r = 3.82$. However, we can also see that the diagonal happens to be very close to intersecting the curve in three different spots. Checking out the graph of the third-iterate map at $r = 3.84$, we can see that these intersections have indeed happened (Figure 10).

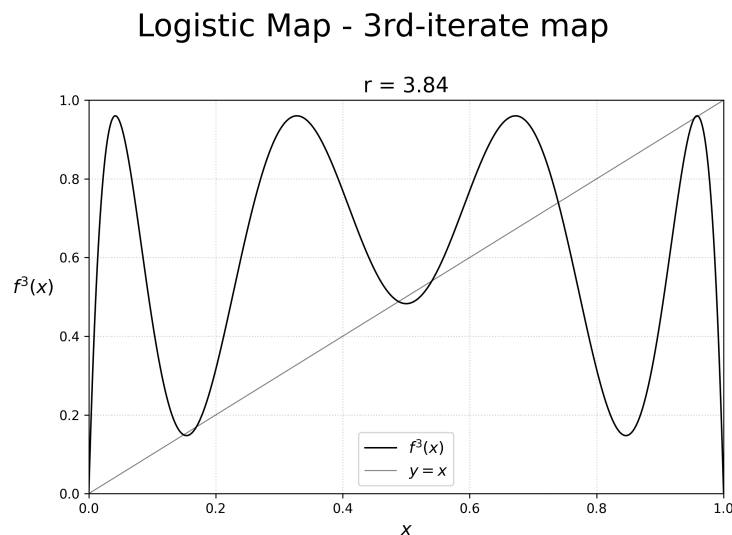


Figure 10: A graph of $f^3(x)$ for $x_{n+1} = 3.84x_n(1 - x_n)$.

At $r = 3.84$, within the 3-cycle window for r , there is now suddenly six more intersection points of $f^3(x)$ with the diagonal. Three of these intersections occur where the slope of $f^3(x)$ is steep. Those three points are unstable fixed points of the third-iterate map, therefore forming an unstable period-3 cycle that would not show up on the orbital diagram. However, the other three of these are where the slope of $f^3(x)$ is shallow ($|\lambda| < 1$), meaning these three points form a stable period-3 cycle which is what is showing up on our orbital diagram.

Now, this tells us that for some r within $3.82 < r < 3.84$, the graph of $f^3(x)$ must have become tangent to the diagonal. Furthermore, we know that at this critical r value the curve becomes tangent to all three hills/valleys at the same time because the intersection points are part of a period-3 cycle that cannot exist with just one or two points. Therefore, the three points must spring into existence at the same time. It is at this critical r that the stable and unstable period-3 cycles are born in a **tangent bifurcation**, which is an analog to the saddle node bifurcations we have studied already in the course. This behavior is what guides the transition from chaos back into periodicity that we see in the orbital diagram.

4 The Lyapunov Exponent and Chaos

While an orbit diagram provides a powerful visual representation of a system's behavior, we also need a rigorous, quantitative metric to determine if a regime is truly chaotic. A defining hallmark of chaos is a sensitive dependence on initial conditions, meaning that two trajectories starting very close to one another will separate exponentially fast over time. To quantify this separation in one-dimensional maps, we use the Lyapunov exponent, denoted by λ (not to be confused with the multiplier, also denoted by λ in previous sections).

To build the intuition for λ , consider an initial condition x_0 and a nearby point $x_0 + \delta_0$, where the initial separation δ_0 is extremely small. We can track how this separation evolves by letting δ_n be the distance between the two orbits after n iterations. If the separation grows (or decays) exponentially, we can model this relationship as:

$$|\delta_n| \approx |\delta_0|e^{n\lambda}.$$

If λ is positive, the separation grows exponentially, signaling sensitive dependence and chaos. If λ is negative, the orbits converge, indicating a stable fixed point or cycle. The more positive or negative λ is, the more chaotic or stable the attractor is.

To derive a computationally useful formula for λ , we can rearrange the exponential growth equation by taking the natural logarithm of both sides :

$$\lambda \approx \frac{1}{n} \ln \left| \frac{\delta_n}{\delta_0} \right|.$$

We can express the separation after n iterates, δ_n , as the difference between the n th iterate of our perturbed initial condition and the n th iterate of our original point:

$$\lambda \approx \frac{1}{n} \ln \left| \frac{f^n(x_0 + \delta_0) - f^n(x_0)}{\delta_0} \right|.$$

We now consider the limit as the initial separation becomes infinitesimally small ($\delta_0 \rightarrow 0$). The term inside the absolute value is precisely the limit definition of the derivative of the n th iterate function, evaluated at x_0 :

$$\lambda \approx \frac{1}{n} \ln |(f^n)'(x_0)|.$$

Evaluating the derivative of $f^n(x_0)$ directly is not feasible for large n . However, applying the chain rule allows us to express the derivative of the n th iterate as the product of the derivatives of the single map evaluated at each point along the orbit:

$$(f^n)'(x_0) = \prod_{i=0}^{n-1} f'(x_i).$$

Substituting this product back into our logarithmic expression, and utilizing the property that the logarithm of a product is the sum of the logarithms, we obtain:

$$\lambda \approx \frac{1}{n} \ln \left| \prod_{i=0}^{n-1} f'(x_i) \right| = \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)|.$$

Finally, to capture the true long-term behavior of the system, we take the limit as the number of iterations n approaches infinity. This yields the formal definition of the Lyapunov exponent for the orbit starting at x_0 :

$$\lambda = \lim_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)| \right).$$

It is important to note that while λ mathematically depends on the specific initial condition x_0 , it evaluates to the same value for all initial conditions that fall within the basin of attraction of a given attractor. Therefore, λ serves as a global property of the attractor itself [4].

4.1 Computing the Lyapunov Exponent

Next we describe a computational method for calculating the Lyapunov exponent given a map $f(x_n)$ and its first derivative $f'(x_n)$, as it is not always possible to do so analytically. For a given r , we first iterate the map from an initial condition long enough for the orbit to settle into its long term behavior. Then, discarding the previous values, we iterate the map another k times for a large k , on each iteration calculating $\ln(f'(x_n))$ and building up a running sum of the logarithms. Finally, we take the running sum and divide it by k , yielding an estimation of λ for that particular r .

Now, if we apply the above method to a range of r values, we can then plot a **Lyapunov graph** of λ with respect to r . It looks like this for the logistic map (Figure 11).

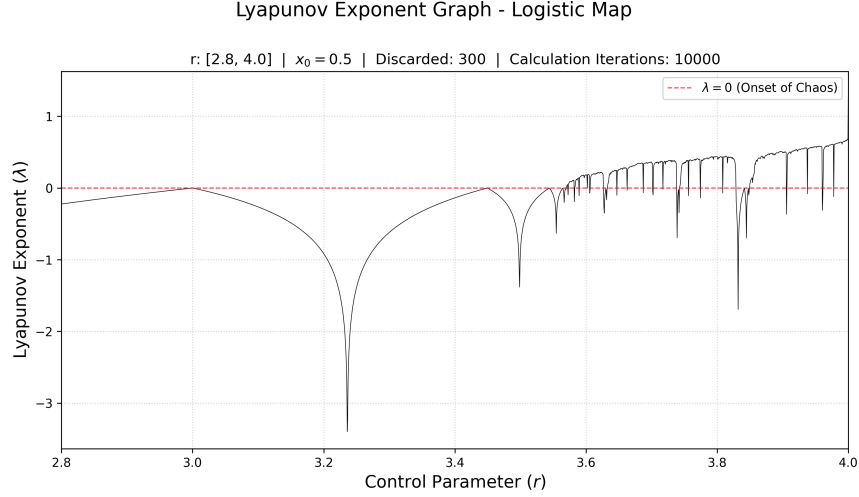


Figure 11: A graph of the Lyapunov exponent λ vs r for $x_{n+1} = rx_n(1 - x_n)$.

Comparing the graph to the orbit diagram, we can confirm the earlier claim of $\lambda < 0$ indicating stability and $\lambda > 0$ indicating chaos. We can see that for $r < r_\infty$, $\lambda < 0$. Beginning at r_∞ the sign of λ then follows the windows of chaotic and periodic behavior visible in the orbit diagram. For example, we see a major dip in λ at the large period-3 window beginning around $r \approx 3.83$ we explored earlier.

5 Universality - Exploring the Ricker Map

Up until now, these results, although fascinating, have been based on solely the logistic map. In order to demonstrate some truly astonishing universal properties of all unimodal maps, we will now explore the Ricker Map, defined by $x_{n+1} = x_n e^{r(1-x_n)}$. Like the logistic map, the Ricker map comes from a context of population biology. Importantly, the Ricker map shares the unimodal structure of the logistic map. As mentioned in the introduction, unimodal maps are maps where $f(x)$ is smooth, concave down, and have a single maximum. The difference now, though, is that whereas the logistic map has a quadratic peak, the Ricker map has a non-polynomial, exponential structure, and generally uses quite a different function from the logistic map. Yet, it will provide striking evidence that many of the properties that we just explored for the logistic map are, in fact, universal.

5.1 Fixed Point Stability Analysis

To begin our analysis of the Ricker map, defined as $f(x) = xe^{r(1-x)}$, we first locate its fixed points by setting $f(x^*) = x^*$:

$$x^* e^{r(1-x^*)} = x^*.$$

This yields two fixed points. The first is the trivial fixed point $x_1^* = 0$. Assuming $x^* \neq 0$, we divide both sides by x^* and take the natural logarithm to find the nontrivial fixed point:

$$e^{r(1-x^*)} = 1 \implies r(1-x^*) = 0 \implies x_2^* = 1.$$

Notice that unlike the logistic map, where the nontrivial fixed point $1 - 1/r$ shifts as the growth rate parameter r changes, the Ricker map's nontrivial equilibrium is rigidly fixed at $x^* = 1$ for all values of r .

To evaluate the stability of these fixed points, we determine the multiplier $\lambda = f'(x^*)$. Applying the product and chain rules to our map yields the derivative:

$$f'(x) = e^{r(1-x)} + xe^{r(1-x)}(-r) = e^{r(1-x)}(1 - rx).$$

For the trivial fixed point $x_1^* = 0$, the multiplier is $\lambda = f'(0) = e^r$. Because $r > 0$ in our biological context, $e^r > 1$, rendering the origin unconditionally **unstable**.

For the nontrivial fixed point $x_2^* = 1$, the multiplier evaluates to:

$$\lambda = f'(1) = e^0(1 - r(1)) = 1 - r.$$

Recalling our linear stability criterion, the fixed point is stable when $|\lambda| < 1$, which gives the inequality:

$$|1 - r| < 1 \implies -1 < 1 - r < 1 \implies 0 < r < 2.$$

Thus, the population will steadily converge to the equilibrium state of $x^* = 1$ as long as $0 < r < 2$. However, exactly at $r = 2$, the multiplier λ reaches -1 . Just as we observed in the logistic map, this marks the threshold of a period-doubling bifurcation. For $r > 2$, the fixed point loses stability, and the orbit is repelled into a period-2 cycle.

5.2 Orbit Diagram and Lyapunov Graph

Next, we'll take a look at the orbit diagram and the Lyapunov graph for the Ricker map (Figures 12 and 13).

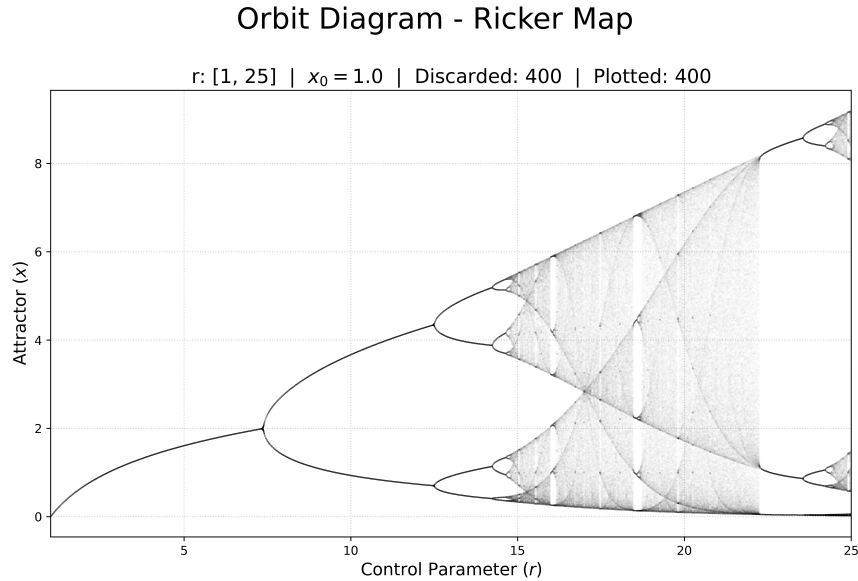


Figure 12: An orbit diagram for $x_{n+1} = x_n e^{r(1-x_n)}$.

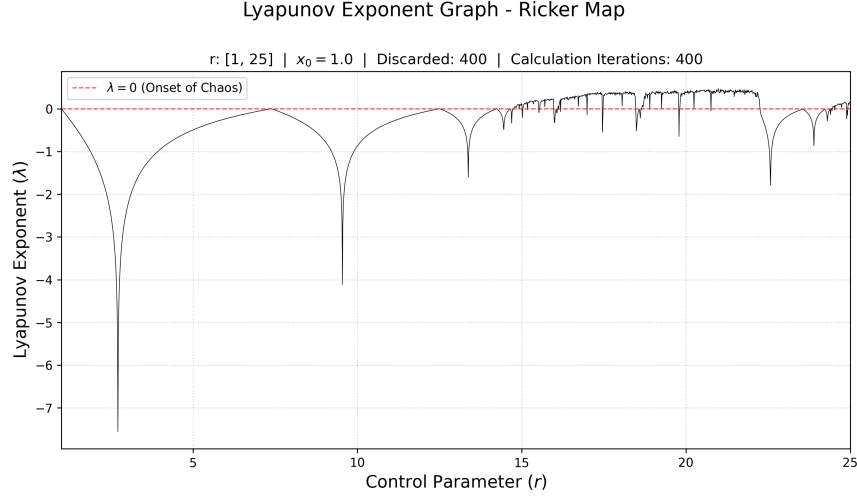


Figure 13: A graph of the Lyapunov exponent λ vs r for $x_{n+1} = x_n e^{r(1-x_n)}$.

The fascinating result is that the orbit diagram for the Ricker map looks identical to that of the logistic map in terms of structure and qualitative behavior, even though there are some quantitative differences with regards to distortion and skew due to the Ricker map using an exponential decay. There is the period doubling route to chaos, which occurs until $r \approx 7.4$, which is where chaos begins for the Ricker map. There is even a large window where a period-3 cycle appears out of the chaos starting at $r \approx 22.3$, similar to the one we explored with the logistic map. Likewise, the Lyapunov graph for the Ricker map matches its corresponding orbit diagram just like with the logistic map.

5.3 Qualitative Universality

In fact, if you were to build up the sequence where each entry is the period of the newly birthed stable cycle as you steadily increase r , and remove any entries greater than 6 (otherwise the sequence would get stuck growing infinitely as $r \rightarrow r_\infty$), you would find that for both the logistic map and the Ricker map, the beginning of the sequence looks like this:

$$1, 2, 2 \times 2, 6, 5, 3, 2 \times 3, 5, 6, 4, 6, 5, 6,$$

where 2×2 indicates a stable cycle birthed from a period doubling (as to differentiate it from stable cycles birthed straight out of chaos) [3]. Though some of the later numbers shown above may be more difficult to see due to the resolution or r range of the orbit diagrams provided, the earlier periods are easily visible. This sequence is known as the U-sequence, and is the same sequence that Metropolis et al. proved to be independent of the unimodal map being iterated (see the original paper for the specific conditions for “unimodal maps” used in the original proof) [3]. The U-sequence is a qualitative form of universality: all unimodal maps follow the same *order* of periodic attractors. But the story doesn’t end there.

5.4 Quantitative Universality

Something that was not mentioned earlier when we looked at the convergence of r to r_∞ for the logistic map is that the convergence is generally geometric. As $n \rightarrow \infty$, the distance between successive period doublings, quantified by $\frac{r_n - r_{n-1}}{r_{n+1} - r_n}$, shrinks by a constant factor of approximately 4.669 [1]. For brevity's sake, we will not derive a way to compute this for the Ricker map as well, but if you were to do so, you would find that $\lim_{n \rightarrow \infty} \frac{r_n - r_{n-1}}{r_{n+1} - r_n} \approx 4.669$ for the Ricker map as well. This is known as Feigenbaum's constant [1, 2], defined as

$$\delta = \lim_{n \rightarrow \infty} \frac{r_n - r_{n-1}}{r_{n+1} - r_n} \approx 4.669.$$

You will find that $\delta \approx 4.669$ for any unimodal map, just as any unimodal map follows the U-sequence.

This is yet another surprising discovery about universality, this time quantitative. Both of these discoveries reveal something extraordinary about the nature of discrete dynamical systems. They demonstrate that the transitions from order to chaos and vice versa are not random and unpredictable, but rather are a highly structured phenomenon that is independent of the specific physical or biological origins of the original models themselves.

6 Conclusion

In this survey, we have explored the rich and surprisingly complex dynamics that arise from simple, one-dimensional iterated maps. Beginning with the foundational tools of fixed point analysis and cobweb diagrams, we traced the logistic map's period-doubling route to chaos. We saw how seemingly erratic behavior can contain hidden structures, such as the stable period-3 window born from a tangent bifurcation, and we rigorously quantified the onset of chaos and sensitive dependence on initial conditions using the Lyapunov exponent.

By extending our computational analysis to the biologically inspired Ricker map, we witnessed the profound principle of universality. Despite their different algebraic foundations—one a quadratic parabola, the other driven by exponential decay—both unimodal maps follow the exact same qualitative U-sequence of periodic attractors and scale quantitatively according to Feigenbaum's constant, $\delta \approx 4.669$. This reveals that the transition to chaos is governed by the fundamental geometry of the map rather than its specific algebraic formulation.

Ultimately, the study of these discrete one-dimensional maps serves as far more than an isolated mathematical curiosity; it provides a vital framework for understanding the continuous dynamical systems we have studied throughout this course. Because the complex dynamics of higher-dimensional continuous flows, such as the strange attractor found in the Lorenz equations, can often be distilled into discrete 1D mappings via Poincaré sections or Lorenz maps [4], the universal laws explored in this paper have sweeping implications. They demonstrate that across diverse scientific disciplines, the chaotic behavior of nonlinear systems is deeply structured and bound by an underlying, universal order.

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All code used to explore computational methods and generate figures can be found on [GitHub](#).

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